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SURVEY

Generative AI for Learning and Teaching Arabic: Opportunities, Challenges, and Ethical Considerations

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ABSTRACT This paper reviews the transformative potential of Generative Artificial Intelligence (GenAI) for Arabic language learning and teaching, with a focus on opportunities, challenges, and ethical considerations. Recognizing that Arabic remains a low-resource language despite its global significance, our methodology was structured around two guiding research questions. For *RQ1 (Current Landscape)*—examining the state of research on GenAI in language education—we conducted a systematic literature search across Scopus, Web of Science, and Linguistics and Language Behavior Abstracts (LLBA). This process retrieved 87 records, from which a small but focused body of studies met the inclusion criteria. The concentration of relevant works underscores both the novelty of the field and the urgent need for foundational research, while ensuring transparency, reproducibility, and rigor in identifying and analyzing the emerging body of relevant work. For *RQ2 (Transferability of Insights)*—investigating what lessons from English language learning can inform Arabic contexts—we employed a purposeful sampling strategy, selecting representative and influential English-language studies from systematic reviews and expert recommendations. This combination balances breadth with depth: the systematic search establishes a reliable evidence base, while purposeful sampling enables focused analysis of best practices in mature domains. Building on these methods, the paper makes three contributions: it delivers a comprehensive synthesis of current trends, applications, and limitations of GenAI in language education; it identifies the unique opportunities and challenges for Arabic, particularly its linguistic complexity and resource scarcity; and it highlights ethical and cultural considerations, including bias, data privacy, and the preservation of linguistic identity. To our knowledge, this is the first survey dedicated to GenAI in Arabic language education, offering a structured foundation for future research, pedagogical innovation, and responsible implementation.

INDEX TERMS Generative AI, language learning, computer-assisted language learning (CALL), arabic language education.

I. INTRODUCTION

Arabic, a major Semitic language with written attestations dating back to the 1st–2nd centuries CE [1], has long shaped culture, knowledge, and communication across vast regions. Its depth is enriched by its role as the language of the Qur'an, giving it profound significance to over two billion Muslims worldwide. Arabic is also a diglossic language, meaning that different varieties are used in different contexts: *Fusha*—which includes both Classical Arabic (CA) and Modern Standard Arabic (MSA)—serves as the high, formal variety used in education, media, and religion, while numerous regional dialects dominate everyday conversation. This functional split, while central to Arabic's vitality, also presents unique challenges for learners. These features make Arabic an especially compelling case for exploring how

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Generative AI (GenAI) can support language learning and teaching in both formal and informal contexts.

Reflecting this linguistic and sociocultural richness, Arabic language learners can be classified into three primary categories—L2 learners, Heritage Learners (HL), and Qara'a Learners (QL)—to better capture the diverse backgrounds, motivations, and learning contexts for which Arabic is acquired. L2 learners are individuals who study Arabic as a second or foreign language—commonly referred to as ASL (Arabic as a Second Language) or AFL (Arabic as a Foreign Language)—often through structured educational programs or immersive environments. Their focus is typically on mastering Arabic for academic, professional, or cultural purposes. Moreover, HL are people who were exposed to Arabic at home or in their communities while growing up in non-Arabic speaking regions. Despite their familiarity with cultural customs and fluency in a local dialect, they frequently have difficulty with *Fusha* registers [2]. Finally, those who initially become proficient in reciting the Quran are known as Qara'a Learners, and they frequently exhibit high phonetic reading abilities but have a limited ability to communicate in the language itself [3]. Creating specialized, culturally aware programs that complement these groups' language and motivational profiles is necessary to meet their demands.

Beyond its historical and cultural significance, Arabic presents both opportunities and challenges in the realm of language learning, particularly given its structural complexity and diglossic nature. Over the years, technology has played a pivotal role in enhancing language learning outcomes, evolving from traditional computer-assisted language learning (CALL) systems to advanced language learning platforms supported by information and communication technology (ICT). The field is now undergoing a significant transformation with the advent of artificial intelligence (AI)—a broad field encompassing machine learning, natural language processing, and more. A particularly transformative subset of AI is GenAI, which focuses on generating human-like content such as text, speech, and images. While AI and GenAI are distinct (with GenAI being a special kind of AI), in this paper we use the terms AI and GenAI interchangeably for simplicity, as our discussion focuses on generative capabilities within the broader AI context. GenAI has the potential to revolutionize how languages, including Arabic, are taught and learned. It introduces capabilities for personalized, interactive, and adaptive learning experiences, addressing language-specific challenges through tailored content generation and real-time feedback [4], [5].

The integration of computer technology into language education began in the late 20th century with early CALL systems [6], [7], which offered multimedia resources and interactive exercises that went beyond traditional classroom methods. The subsequent rise of the internet and ICT expanded this shift, leading to widespread use of e-learning platforms and language apps that broadened access to language education.

GenAI holds significant potential to enhance language learning by strengthening both receptive and productive skills. It can support receptive skills through interactive listening and reading comprehension tools and aid productive skills with speech synthesis and writing assistance. For instance, GenAI chatbots can engage learners in spoken dialogue for listening and speaking practice, while text-generation models can aid reading comprehension and provide writing support, particularly for second-language learners.

TABLE 1. Integration of language skills with common european framework of reference (CEFR) language modes.

Mode	Skills
Receptive (Reception)	Listening, Reading
Productive (Production)	Speaking, Writing
Interaction	Speaking and Listening, Reading and Writing
Mediation	Translating and Interpreting, Summarizing and Adapting

TABLE 2. Comparison of our paper with the existing papers. (Opp. = Opportunities, Chal. = Challenges, Ethics = Ethical aspects, Culture = Cultural aspects.)

Paper	Year	Language	Application	Opp.	Chal.	Ethics	Culture
[13]	2024	Multilingual	Language learning	✓	✓	✓	✓
[14]	2023	English	Listening, speaking, reading, writing	✓	✓	–	–
[15]	2024	English	Language teaching and learning	✓	✓	✓	–
[16]	2023	Multilingual	Language learning	✓	✓	✓	–
[17]	2023	Indonesian	Academic writing	✓	–	–	–
[18]	2024	English	Academic communication	✓	✓	✓	–
[19]	2024	Multilingual	Listening	✓	✓	✓	–
Ours	2025	Arabic	Language teaching and learning	✓	✓	✓	✓

While AI-driven tools have accelerated English language learning—thanks to robust datasets and substantial research investments—similar progress for Arabic remains limited. The nuanced structure of Arabic, marked by its diverse dialects and relatively scarce data, presents unique challenges that must be addressed to leverage the full potential of GenAI for Arabic learners [8], [9].

Moreover, the evolution from "four skills"—*listening, reading, writing, and speaking*—to "four modes" of language learning, as proposed by the Common European Framework of Reference for Languages Companion Volume (CEFR-CV), represents a shift toward context-driven language use. Traditionally, the four language skills are categorized into receptive and productive skills: listening and reading are receptive skills, involving the intake and comprehension of information, while speaking and writing are productive skills, requiring the creation and articulation of thoughts and ideas [10]. The four modes of language learning proposed by CEFR-CV—*production, reception, interaction, and mediation*—emphasize active engagement in real-world communication. This approach is especially beneficial for Arabic learners, enabling them to navigate linguistic and cultural nuances effectively through dynamic, situational language use rather than solely grammar-based methods [10], [11], [12]. Table 1 shows how traditional language skills are linked with the CEFR language modes.

This survey paper explores the transformative potential of GenAI in Arabic language learning, providing a

comprehensive review of how insights from English language education can inform Arabic instruction. As a survey, it synthesizes trends, gaps, and challenges in the field rather than conducting empirical evaluations or in-depth technical analyses. It examines unique opportunities and challenges, alongside ethical considerations, to guide future research and applications. Table 2 summarizes related work, highlighting contributions and differences from this paper. To our knowledge, this is the first survey to address GenAI in Arabic language teaching and learning, emphasizing opportunities, challenges, and ethical considerations within a cultural context. While related work exists for English and other languages, this paper fills a critical gap by focusing on Arabic.

The rest of the paper is organized as follows: Section II reviews technology's role in language learning; Section III highlights the methodology used to answer our main research questions; Section IV draws insights from English education for Arabic; Section V explores GenAI opportunities and challenges in Arabic teaching; Section VI discusses ethical considerations, open issues and future directions. Section VII concludes the paper.

II. TECHNOLOGY AND LANGUAGE LEARNING

Technology has significantly influenced the evolution of language learning, beginning with the early use of computer-assisted language learning tools, which provided learners with interactive ways to practice and improve their language skills. As the internet became more accessible, language learning expanded to online platforms, offering greater flexibility and a wider range of resources. Today, the focus is shifting toward AI, which is transforming the landscape of language learning. This section will examine the journey from traditional technology-based language learning methods to the integration of AI, highlighting its growing role and potential to reshape the way languages are taught and learned in the digital age.

A. HISTORIC CONTEXT AND CURRENT RELEVANCE

1) EARLY COMPUTER-ASSISTED LANGUAGE LEARNING SYSTEMS

For Arabic, exploratory CALL (Computer-Assisted Language Learning) initiatives began as early as the 1960s–70s, using mainframe computing, machine language programming, and punch card inputs. These pioneering efforts expanded in the 1980s with the advent of personal computers and user-friendly software [20]. Progress was often driven by collaborations between Western scholars and Arab researchers, resulting in practical tools such as Arabic fonts for Apple and CALL software templates. By the mid-1980s, the Arab world was increasingly shaping standards for Arabic's electronic representation, contributing to milestones like Unicode and UTF-8 [20]. More broadly, CALL systems—as discussed in the wider literature—introduced interactive and independent ways of engaging with language materials and laid the groundwork for integrating multimedia and software tools into education [6].

2) THE IMPACT OF ICT

The rise of ICT further expanded the possibilities for language learning, providing more accessible, flexible, and resource-rich environments for learners [21]. ICT tools facilitated e-learning platforms and online courses, making language education more widely available across different regions [22].

3) ONLINE LANGUAGE LEARNING COMMUNITIES

The Internet has enabled the formation of online language learning communities, where learners can practice with native speakers and access diverse language resources [23]. These communities support the acquisition of speaking and writing skills by allowing learners to engage in authentic language use.

4) E-LEARNING PLATFORMS AND MOBILE APPLICATIONS

E-learning platforms and mobile apps have been used as popular tools for language education, offering flexibility and a range of interactive features even before the uptake of modern GenAI technology [24]. More recently, massive open online courses (MOOCs)—and particularly Language MOOCs (LMOOCs) designed for language acquisition—have expanded these opportunities, providing scalable access to language learning worldwide [25]. Research shows that such platforms can significantly enhance language acquisition through multimedia, gamification [26], and adaptive features [27].

5) AI-POWERED TOOLS FOR PERSONALIZED LEARNING

AI has introduced new capabilities for delivering personalized language learning experiences by providing instant feedback and adapting to individual learners' needs [15]. The integration of AI with traditional language learning approaches has led to improved learner engagement and outcomes. Moreover, Figure 1 illustrates the primary computer-based language teaching methods.

B. GenAI AND LANGUAGE MODELS

GenAI, particularly LLMs, represents a significant advancement in language education by enabling automated text generation, conversational practice, and grammar correction [4]. Such models are capable of generating human-like responses and simulating language use in various contexts [8]. The evolution of AI in language learning has led to significant advancements, particularly with the introduction of GenAI and LLMs. Earlier models, such as Naïve Bayes, Support Vector Machines (SVM), and Recurrent Neural Networks (RNNs), laid the foundation for Natural Language Processing (NLP) applications. While these classical models were instrumental in early text classification and sentiment analysis tasks, they faced limitations in contextual understanding



FIGURE 1. Taxonomy of computer-based language teaching. The figure categorizes methods such as CALL, e-learning platforms, mobile apps, AI, and internet-based communities, with specific examples of tools and techniques [28].



FIGURE 2. Taxonomy of GenAI technology for teaching and learning arabic informed by existing reviews on AI in English L2 contexts [29], [30]. Categories such as personalized learning, conversational AI, writing assistance, and ethical considerations are grounded in prior studies, while dialectal processing and orthographic control are added to reflect challenges unique to Arabic [31].

and adaptability [32]. The emergence of deep learning models, such as Long Short-Term Memory (LSTM) networks and Transformer-based architectures, revolutionized the field by enabling more sophisticated language processing. Transformer models introduced self-attention mechanisms, allowing for parallel sequence processing and improved contextual comprehension [33]. This shift set the stage for the development of advanced GenAI models like GPT, BERT, and multimodal architectures, which now power interactive, personalized, and adaptive language learning experiences.

To provide a structured comparison of these advancements, Table 3 highlights the strengths, weaknesses, and applications of various classical NLP models, demonstrating their contributions and limitations in language learning. Moving beyond these traditional methods, Table 4 outlines the evolution of Transformer-based architectures, detailing their impact on modern language learning and their application in Arabic NLP. Additionally, as training techniques play a crucial role in enhancing AI models, Table 5 presents a comparative analysis of different training methodologies, including pre-training, fine-tuning, transfer learning, and reinforcement learning with human feedback. Through bridging the gap between foundational NLP techniques and state-of-the-art GenAI solutions, this section emphasizes how these advancements have transformed language learning. The integration of generative models not only enhances accessibility and engagement but also opens new pedagogical possibilities, setting the stage for more immersive and personalized educational experiences.

By tracing the evolution of technology in language learning, this section provides essential contexts for all readers, including those without an in-depth understanding of the field. It underscores the growing influence of AI-driven tools in shaping modern educational practices. As traditional methods give way to more adaptive and intelligent systems, the integration of GenAI and LLMs marks a pivotal shift in how languages are taught, learned, and personalized to individual needs. This progression not only enhances accessibility and engagement but also introduces new pedagogical possibilities. To ensure that this paper is accessible to a broad audience, we present this overarching introduction before delving into more specialized discussions. Finally, Table 6 compares traditional classroom learning, technology-assisted learning, traditional AI-driven learning, and GenAI-driven learning.

C. GenAI AND TEACHING ARABIC LANGUAGE

Arabic's diglossic nature, intersecting with its multi-dialectical reality and deep cultural and religious ties, creates unique learning motivations and challenges [39]. GenAI can address these by customizing learning experiences that balance both formal and informal Arabic across contexts. Through dialect-specific simulations, GenAI enables learners to engage in realistic dialogues that reflect regional nuances, such as Egyptian, Levantine, or Gulf Arabic. By blending *Fusha* with colloquial elements in professional, social, or religious scenarios, GenAI fosters functional fluency and cultural competence. This adaptable approach meets learners' specific goals, making GenAI an invaluable tool for navigating the complexities of Arabic language instruction

TABLE 3. Comparison of traditional AI/ NLP models with the transformer model.

Model	Description	Strengths	Limitations	Applications
Naïve Bayes [34]	A probabilistic classifier based on Bayes' theorem, assuming independence among features.	Simple, fast, effective for text classification.	Assumes feature independence, which may not be held in complex language tasks.	Text classification, spam filtering, sentiment analysis.
Support Vector Machine (SVM) [35]	A model that finds the optimal hyperplane separating different classes in the feature space.	Effective in high-dimensional spaces, robust to overfitting.	Computationally intensive with large datasets, less effective with noisy data.	Text categorization, sentiment analysis, named entity recognition.
Recurrent Neural Network (RNN) [36]	A neural network model designed for sequential data, capturing temporal dependencies.	Suitable for tasks like language modeling and sequence prediction.	Prone to vanishing gradient problem, making training challenging.	Language modeling, machine translation, speech recognition.
Long Short-Term Memory (LSTM) [37]	An extension of RNNs that addresses the vanishing gradient problem, enabling learning of long-term dependencies.	Effective for tasks requiring understanding of context over long sequences.	Computationally intensive requires significant resources.	Machine translation, speech processing, text generation.
Transformer Models [38]	Models utilize self-attention mechanisms to process sequences in parallel, capturing complex relationships.	State-of-the-art performance in various NLP tasks, pre-trained on large corpora.	High computational requirements, large model sizes.	Text generation, machine translation, question answering.

in a culturally sensitive and engaging way. Finally, Figure 2 summarizes the areas of GenAI for Arabic teaching and learning.

This paper seeks to address two key research questions (RQs):

1) **RQ1**: What has been explored in the connection between GenAI and Arabic as a second language?
2) **RQ2**: What major insights can research on GenAI and English as a second language (ESL) offer for Arabic?

By investigating existing work on Arabic L2 learning with GenAI and analyzing transferable lessons from English L2 contexts, the paper aims to identify opportunities, gaps, and best practices for integrating GenAI into Arabic language education.

III. METHODOLOGY

The methodology was designed to directly address the study's two central research questions. First, to explore RQ1, a systematic search was conducted to identify and analyze the limited but emerging body of work in this area. Second, to address RQ2, we did not conduct a full systematic review of this vast English-language literature but instead adopted a purposeful sampling approach. We focused on representative and influential studies drawn from systematic reviews, and expert recommendations from co-authors that examine major GenAI. The goal was to extract the most relevant insights from English L2 learning and assess their transferability to Arabic. Together, these approaches provide a structured framework for mapping existing knowledge and identifying gaps that future research can address. Purposeful sampling for RQ2 was continued until conceptual saturation was reached, defined as the point at which additional English-language review studies no longer yielded new categories of GenAI applications or transferable pedagogical insights.

A. EXPLORING THE USE OF GenAI IN TEACHING ARABIC AS A SECOND LANGUAGE

Although this paper is structured as a narrative review, to address the first research question, we employed a systematic search strategy consistent with PRISMA guidelines to explore the existing literature. The aim was to identify and evaluate research on the application of GenAI—particularly large language models (LLMs) and chatbot-based systems—in Arabic language education targeting second language (L2), foreign language, or heritage language learners. Each database was queried using carefully constructed Boolean search strings and field-specific syntax aligned with its indexing structure. In Scopus, we used the TITLE-ABS-KEY field to search for combinations of terms such as "generative AI", "large language model*", "ChatGPT", "Claude", "Gemini", "LLaMA", and "prompt engineering", intersected with Arabic language learning terms like "Arabic", "second language", "L2", "foreign language", "language learner*", and "language acquisition". In Web of Science, the query included terms such as "artificial intelligence", "machine learning", "automated essay scoring", and "AI writing tool" combined with "Arabic" and various second-language learning descriptors using AND/OR operators. For LLBA, we performed a comprehensive search using the Ti (title), Ab (abstract), and Su (subject) fields, employing a broad set of GenAI-related terms (e.g., "GPT-3.5", "Gemini Pro", "instruction tuning", "AI chatbot*", "natural language generation") along with Arabic language learning descriptors to ensure high recall. These database-specific strategies enabled us to retrieve a wide scope of potentially relevant literature, reflecting the interdisciplinary and rapidly evolving nature of the GenAI field.

A total of 87 records were retrieved. After removing three duplicate entries, 84 records were screened. Only four studies satisfied our inclusion criteria. Of these, two articles directly investigated the implementation or assessment of GenAI tools in Arabic language teaching, demonstrating clear alignment with our focus. The remaining two papers only partially met our criteria, addressing GenAI at a conceptual or thematic level without detailed empirical investigation. The limited number of qualifying studies highlights the early stage of scholarly inquiry in this domain. Despite the transformative potential of GenAI in language education, its application to Arabic—a linguistically rich but underrepresented language in NLP research—remains

TABLE 4. Comparison of transformer-based architectures for language learning and arabic NLP applications.

Architecture	Transformer-Based Features	Strengths for Language Learning	Limitations	Applications in Language Learning	Applied to Arabic Language
Transformer Foundational [38]	Processes sequences in parallel using self-attention mechanisms.	Foundational for capturing long-range dependencies in text.	High computational demands, not task specific.	Backbone for all modern NLP models; essential for machine translation and text generation.	Yes, as a foundational architecture for Arabic NLP models.
GPT (Generative Pre-trained Transformer) [40]	Autoregressive Transformer variant for generating coherent, contextually rich text.	Excels in generative tasks like text generation and dialogue systems.	Prone to bias, requires large datasets, struggles with factual accuracy.	Interactive language tutors, vocabulary-building, creative writing, and conversational agents.	Limited: Some Arabic text generation capabilities exist but lack extensive fine-tuning for Arabic.
AraGPT2 [41]	Arabic-specific adaptation of GPT-2 trained on Arabic text corpora.	Optimized for Arabic text generation, better coherence in Arabic than general GPT models.	May still carry biases from training data, lacks deep contextual understanding like BERT.	Arabic language tutoring, storytelling, conversational agents, creative writing assistance.	Yes, specifically designed for Arabic NLP.
BERT (Bidirectional Encoder Representations from Transformers) [42]	Bidirectional Transformer focuses on understanding relationships between words in context.	Strong contextual understanding for comprehension and analysis tasks.	Not designed for generative tasks; requires task-specific fine-tuning.	Grammar correction, sentence understanding, reading comprehension tools.	Yes, adapted for Arabic (e.g., AraBERT).
T5 (Text-to-Text Transfer Transformer) [43]	Unified framework treating all NLP tasks as text-to-text problems.	Versatile across language tasks, simplifies model usage.	Computationally expensive, fine-tuning needed for specific tasks.	Summarization, translation, question generation, and grammar correction.	Limited: mT5 includes Arabic but performance can be improved with additional fine-tuning.
Fanar Star and Fanar Prime [44]	Arabic-specific Transformer-based models trained on large, multilingual, and multimodal datasets.	Tailored for Arabic NLP, robust handling of dialects and script complexities.	Limited by availability of high-quality Arabic-specific datasets.	Arabic grammar exercises, dialect translation, text-to-speech for Arabic learners, speech-based language practice.	Yes, explicitly designed for Arabic applications.
BART (Bidirectional and Auto-Regressive Transformers) [45]	Combines BERT's understanding with GPT's generation capabilities.	Strong for tasks requiring both understanding and generation.	Computationally expensive, requires extensive pre-training for new tasks.	Text summarization, question generation, paraphrasing tools for comprehension.	Limited: No Arabic-specific variant, but adaptable through multilingual pre-training (e.g., mBART).
mT5 (Multilingual T5) [46]	Multilingual extension of T5, supporting over 100 languages.	Supports low-resource languages and cross-lingual tasks.	Larger and slower than T5; limited performance in underrepresented languages.	Multilingual platforms, cross-lingual exercises, and low-resource language translation.	Yes, Arabic is included in mT5 but may require more fine-tuning for dialectal and script-specific nuances.
Whisper [47]	Transformer-based model for robust automatic speech recognition (ASR).	High transcription accuracy, robust in noisy environments.	High computational demands, slower inference for real-time applications.	Speech-to-text for pronunciation, listening comprehension, and dictation tasks.	Yes, supports Arabic for speech recognition.

relatively unexplored. This points to a significant research gap and an opportunity for further empirical and theoretical work to guide the responsible and effective integration of GenAI in Arabic language pedagogy. The selected papers are briefly described in Section IV-C and are summarized in Table 7.

Studies were included if they explicitly examined Generative AI (e.g., large language models or chatbot-based systems) in the context of Arabic language learning or teaching for second, foreign, or heritage language learners and provided empirical evaluation or concrete pedagogical implementation. Studies were excluded if they focused solely on Arabic NLP without a pedagogical component, addressed AI in general rather than GenAI specifically, were purely conceptual or opinion-based, duplicated prior records, or lacked accessible full text.

The large initial yield of 87 records reflected our intentionally broad search strategy to capture any potentially relevant work at the intersection of AI and Arabic language teaching. After title, abstract, and full-text screening, only four articles met all inclusion criteria. Most excluded studies were either not focused on Arabic language learning or teaching, addressing Arabic NLP or speech recognition without a pedagogical context; did not involve GenAI, instead covering AI in general or other computational tools; were purely conceptual or opinion pieces without empirical data or evaluation; were duplicates indexed across multiple sources; or were not in English or had inaccessible full text. This pattern highlights the limited but emerging scholarly work directly examining the intersection of GenAI and Arabic language pedagogy.

B. LEVERAGING ENGLISH L2 RESEARCH TO INFORM ARABIC L2 CONTEXTS

Since English is the most commonly used resource-rich language in the world, it provides a solid foundation for comprehending how GenAI is used in language instruction. It is highly suited for training sophisticated AI models because of its wide digital presence, ample and varied dataset availability, and well-documented language frameworks. Because of these qualities, English is used as the primary training language for the majority of GenAI systems. As a result, English is now a crucial component for evaluating the opportunities and limitations of GenAI in education. While other high-resource languages, such as Spanish and French, contribute valuable data to AI development [54], they remain secondary to English in terms of their influence on AI tools and their widespread adoption in language learning systems.

1) SPEAKING AND PRONUNCIATION

A systematic review by Wang, Abdul Aziz, and Kutty [55] analyzed 21 studies on integrating AI into EFL speaking

TABLE 5. Comparative analysis of NLP training techniques for language learning in arabic.

Training Technique	Description	Strengths	Limitations	Language Learning Applications	Applied to Arabic
Pre-training & Fine-tuning [48]	Pre-train on large corpora, then fine-tune for specific tasks or domains.	Strong task performance; adaptable to domain needs.	Data- and compute-intensive.	Grammar correction, vocabulary learning, task-specific tutors.	Yes; used in AraBERT, Fanar, and mT5.
Transfer Learning [49]	Adapting pre-trained models to new tasks with limited additional training.	Efficient; well-suited for low-resource settings.	Dependent on source–target similarity.	Cross-lingual learning, dialect adaptation, translation.	Yes; widely used in Arabic NLP (e.g., mT5, AraBERT).
Few-/Zero-shot Learning [50]	Perform tasks with few or no task-specific examples.	Highly flexible; minimal data requirements.	Lower accuracy on complex tasks.	Rapid task adaptation, multilingual and dialectal learning.	Limited; inconsistent performance for Arabic.
RLHF [51]	Fine-tuning using human feedback to guide model behavior.	Improves alignment with pedagogical and ethical goals.	Resource-intensive; requires expert feedback.	Personalized tutoring, ethical and cultural adaptation.	Emerging; limited Arabic-specific adoption.
Multimodal Pre-training [52]	Joint training on text, image, and audio data.	Supports richer, multimodal learning interactions.	High data and compute requirements.	Pronunciation support, visual aids, interactive learning.	Limited; partial support in Fanar and Whisper.
Active Learning [53]	Iterative training using the most informative samples.	Reduces labeling effort; improves efficiency.	Slower training; requires expert oversight.	Adaptive vocabulary and grammar learning.	Emerging; promising for dialect adaptation.

TABLE 6. Comparison between different language learning approaches.

Method	Description	Strengths	Weaknesses	Relevance to Arabic Language Education
Traditional Classroom Learning	Teacher-led, in-person instruction with a focus on grammar, vocabulary, and conversation.	Interactive and personalized feedback from teachers; culturally and contextually relevant.	Limited scalability; dependent on teacher expertise; less adaptable to individual learning speeds.	Effective for foundational learning but struggles with dialect diversity and script complexities.
Technology-Assisted Learning	Use of multimedia tools (e.g., apps, videos, flashcards) to supplement instruction.	Engaging visuals and interactive aids; scalable across larger audiences.	Limited personalization; lacks advanced contextual understanding of Arabic dialects and nuances.	Useful for vocabulary building, but cannot address advanced grammatical or contextual learning needs.
Traditional AI-Driven Learning	Rule-based systems, static NLP models (e.g., Naïve Bayes, SVMs, or early RNNs).	Task-specific solutions like grammar correction, basic text-to-speech; scalable and reliable.	Limited adaptability, lacks contextual awareness, and struggles with understanding dialect diversity.	Useful for basic tasks (e.g., grammar correction, simple text analysis) but limited for Arabic dialects.
GenAI-Driven Learning	Dynamic, context-aware AI using models like GPT, BERT, Fanar, and Whisper.	Highly personalized, scalable, supports multimodal learning (text, audio, images); context-aware.	Requires high-quality data and computational resources; may generate biased or inconsistent outputs.	Effective for addressing script complexity, dialect variations, and advanced personalized learning.

instruction in Asian higher education. The review found a significant increase in research since 2022, primarily in East Asia, with a focus on quantitative studies. AI tools such as chatbots, apps, and virtual human teachers were identified as prevalent in EFL speaking instruction, with task-based learning emerging as the most effective instructional method. However, challenges such as technological deficiencies, emotional demotivation, learner variability, lack of AI literacy, teaching-related issues, and overreliance on AI were noted as barriers to full adoption.

In the context of Arabic-speaking learners, the integration of AI in speaking instruction could offer personalized practice opportunities, particularly in task-based learning scenarios. However, challenges such as technological deficiencies, emotional demotivation, learner variability, lack of AI literacy, teaching-related issues, and overreliance on AI may also hinder the effective implementation of AI in Arabic-speaking EFL contexts.

2) WRITING AND AUTOMATED WRITING EVALUATION

Feng, Li, and Zhang [29] reviewed 112 studies from 2014 to 2024 to investigate how AI supports second language writing. The findings highlighted that most research focused on English learners at the university level, exploring AI-driven tools such as Automated Writing Evaluation systems, grammar checkers, and intelligent tutoring platforms. These technologies were shown to improve learners' writing quality, foster independence, and promote more effective revision practices. Nevertheless, issues such as excessive dependence on AI, insufficient digital literacy, and potential privacy risks were commonly reported [29]. For Arabic-speaking learners, similar AI tools could offer tailored support, enhancing writing development. However, the unique characteristics of Arabic, including its right-to-left script, and rich morphology may pose additional challenges for AI integration, requiring specialized adaptation of these technologies to ensure accurate and effective feedback.

3) CHATBOTS AND TEACHER ROLES

Li, Zhou, and Chiu [56] conducted a systematic review focusing on the roles of teachers in AI chatbot-assisted language learning, framed through Self-Determination Theory (SDT). The study identified that teachers act as facilitators, motivators, and scaffolding providers, enhancing learners'

TABLE 7. Summary of the selected papers drawn from the systematic search.

Authors(Year)	Title	GenAI Tool/Type	Language Focus	Learner Type	Study Focus	Publication Type	Source
Alkaabi, M. Almaamari, A. (2025)	Generative AI Implementation and Assessment in Arabic Language Teaching.	Generative AI (LLM, unspecified)	General Arabic Teaching	L2 learners	Implementation and assessment	Journal Article	IJOPCD
Allaithy, A. and Zaki, M. (2025)	Evaluation of AI-generated reading comprehension materials for Arabic language teaching.	AI-generated materials (likely LLM-based)	Reading comprehension	L2 learners	Material evaluation	Journal Article	CALL
Hanandeh, A. et al. (2024)	Artificial Intelligence in Arabic Linguistic Landscape: Opportunities, Challenges, and Future Directions.	General AI (discusses AI contextually)	Linguistic landscape	Mixed/general	Material Thematic review	Journal Article	TEM Journal
Issa, E. and Hammond, M. (2023)	KalaamBot and KalimaBot: Applications of chatbots in learning Arabic as a foreign language.	Chatbots (KalaamBot, KalimaBot)	Arabic as a foreign language	Foreign language learner	Application of chatbots	Book chapter	IGI global

autonomy, competence, and relatedness. However, the review also highlighted challenges such as teachers' limited AI literacy and the potential for reduced interpersonal interactions.

Meniado [30] explored the impact of ChatGPT on English language teaching, learning, and assessment. The study found that ChatGPT can support language learning by providing meaning-focused inputs, offering guidance during output production, giving feedback on accuracy, and facilitating fluency development. However, issues such as inaccurate responses, academic dishonesty, and privacy concerns were also noted.

For Arabic-speaking learners, integrating AI chatbots like ChatGPT into language learning presents both opportunities and challenges. The complex nature of the Arabic language may pose difficulties for AI systems in providing accurate feedback. Also, teachers' limited AI literacy and the potential for reduced interpersonal interactions may impact the effectiveness of AI integration in Arabic language learning.

IV. INSIGHTS FROM ENGLISH LANGUAGE LEARNING AND IMPLICATIONS FOR ARABIC

A. AI APPLICATIONS FOR ENGLISH LANGUAGE LEARNING

AI-based tools for grammar correction and writing assistance have shown effectiveness in supporting English language learners, especially for ESL contexts [57]. These tools can provide real-time feedback and suggestions for improving writing quality [58]. For instance, ProWritingAid offers comprehensive insights into writing coherence and clarity, while Grammarly provides contextual recommendations to improve grammar and style. In the same way, LanguageTool is a flexible option for ESL learners because it enables multilingual users. These platforms promote a deeper comprehension of linguistic rules by not only pointing out mistakes but also outlining fixes.

Conversational agents, such as chatbots powered by GenAI, can serve as language partners for learners, offering spoken language practice in a risk-free environment [59]. These tools have been shown to improve learners' speaking abilities by providing immediate feedback and language practice opportunities. For example, Duolingo's AI-powered bots mimic engaging dialogues catered to learners' skill levels, while ChatGPT enables open-ended discussions. Platforms such as Rosetta Stone also use speech recognition technology to assess fluency and pronunciation, which improves speaking abilities in real time.

AI-powered reading comprehension tools are transforming the way students interact with texts by offering customized assistance in comprehending content and context. By tailoring reading materials to each learner's level of expertise, platforms such as ReadTheory provide questions that strengthen comprehension and critical thinking abilities. These resources use NLP to assess the difficulty of texts and produce tests, summaries, and vocabulary expansions, assisting students in gradually increasing their comprehension [60]. Similar technologies can help learners of Arabic by giving cultural context for informal idioms, simplifying complicated texts, and creating context-relevant definitions. Real-time feedback and the exploration of nuances in formal and informal Arabic texts are two more benefits of integrating such tools with GenAI models.

B. LESSONS LEARNED FROM ENGLISH LANGUAGE EDUCATION

The success of AI-driven tools in English language education highlights valuable strategies that can be adapted to Arabic language learning. Adaptive learning algorithms can help personalize content and feedback, catering to individual learner needs and fostering autonomy by dynamically adjusting to different proficiency levels [70]. This personalized approach ensures that learners progress at their own pace, addressing specific challenges that arise from Arabic's rich morphology and syntactic complexity. Multimedia approaches have further revolutionized English learning by integrating text, audio, video, and images to create immersive, multisensory experiences. These methods cater to diverse learning styles—auditory learners benefit from podcasts, while visual learners engage with videos and interactive exercises. Combining auditory and visual stimuli enhances cognitive processing and retention, bridging the gap between theoretical knowledge and practical application [71]. Applying these approaches to

TABLE 8. Existing AI-based tools for language learning.

Tool	Skill Targeted	Key Functions	Accessibility	Open Source	Target Audience
Grammarly [61]	Writing	• Rewrite paragraphs • Provide grammar suggestions • Punctuation, spelling, tone, and plagiarism detection.	Desktop, web browser, mobile devices.	No	General individuals, professionals, and students.
LanguageTool [62]	Writing	• Grammar style and spell check • Multilingual	API, web browser, desktop and mobile applications.	Yes	General individuals, professionals, and students.
ProWritingAid [63]	Writing	• Grammar style and spell check • Story critique • Rephrase suggestions • Writing analysis report	Desktop and web browser.	No	Students, authors, writers, and professionals.
ChatGPT [64]	Writing, speaking, reading, and listening	• Conversational AI • Offered in multiple languages • Customizable prompts • Supports multimedia • Remembers contexts within conversations (memory)	Web browser, desktop, and mobile applications.	No	Diverse range of individuals and groups.
Duolingo [65]	Writing, speaking, reading, and listening	• Gamified learning experience • Personalized learning using AI • Diverse language offerings • Progress tracking	Web browser, desktop, and mobile applications.	No	Beginners and casual learners.
Babbel [66]	Speaking and listening	• Immersive language lessons • Live online classes • Personalized learning with a focus on conversational skills and cultural knowledge	Web browsers and mobile applications.	No	Travelers, and casual learners.
Rosetta Stone [67]	Speaking and listening	• Pronunciation aid using speech recognition • Online and offline immersive language lessons • Personalized learning with a focus on conversational skills and cultural knowledge	Web browser, desktop, and mobile applications.	No	General learners, students, or enterprises.
ReadTheory [68]	Reading	• Focuses on enhancing reading comprehension • Adaptive to students' levels • interactive reading experience	Web browsers.	No	Students (K-12).
Speechling [69]	Speaking and listening	• Applies a combination of technology and human coaching to help students improve their speaking skills • Supports multiple languages	Web browsers and mobile applications.	No	General learners.

Arabic education can provide learners with contextualized content, making the language more accessible and engaging, and supporting deeper mastery of its orthographic and phonological nuances.

C. GenAI AND ARABIC LANGUAGE LEARNING

The application of GenAI in Arabic language learning is inherently tied to advancements in AI model development. A summary of some of the existing AI-based language learning tools is displayed in Table 8. Recent breakthroughs in LLMs have introduced enhanced contextual understanding and improved multimodal capabilities, significantly contributing to AI-driven Arabic learning tools. However, challenges such as the limited availability of high-quality Arabic datasets, the diversity of Arabic dialects, and the structural complexity of the Arabic script remain significant barriers. A notable effort addressing these challenges is the Fanar platform, an Arabic-centric multimodal GenAI ecosystem. Fanar's advanced LLMs—Fanar Star and Fanar Prime—have been specifically designed to support Arabic-language applications by leveraging an extensive dataset of approximately 1 trillion Arabic, English, and code tokens. These models demonstrate state-of-the-art performance on Arabic NLP benchmarks and provide robust capabilities for Arabic text generation, speech recognition, and multimodal tasks [44]. Their development marks a significant step toward improving the accessibility, accuracy, and versatility of AI-driven Arabic language learning tools. As GenAI continues to evolve, advancements like these will play a transformative role in Arabic language education. By addressing linguistic and technological challenges, such tools can introduce new pedagogical possibilities, personalized learning experiences, and innovative ways to engage learners across diverse contexts.

To better understand the current state of research, we analyzed four studies that met our inclusion criteria in Section III-A. Their findings provide initial evidence of how GenAI is being explored in Arabic L2 education across key skills such as writing, reading, and conversational practice.

1) GenAI AND ARABIC L2 WRITING AND PEDAGOGICAL SUPPORT

Alkaabi and Almaamari [72] conducted an empirical qualitative investigation into how Arabic language instructors in the UAE use GenAI tools such as ChatGPT. Through in-depth interviews, the study documents pedagogical applications including lesson planning, writing assistance, and formative assessment. Notably, the study provides insights into the challenges faced by educators—such as limitations in dialectal support and concerns over the trustworthiness of AI outputs. These findings highlight the importance of context-sensitive deployment of GenAI in real-world classroom environments and make a meaningful contribution to the limited empirical literature in this area.

2) GenAI AND ARABIC L2 READING COMPREHENSION

Allaithy and Zaki [73] offer one of the few systematic evaluations of AI-generated educational content in Arabic. Their study compares the outputs of five GenAI platforms in terms of syntactic correctness, semantic coherence, and pedagogical appropriateness in reading comprehension exercises. The analysis reveals that while the materials are diverse and often engaging, they also exhibit translation artifacts, cultural mismatches, and occasional syntactic anomalies. The authors advocate for human oversight in content deployment, emphasizing the need for educator curation to harness GenAI's potential while mitigating its limitations.

3) GenAI AND ARABIC L2 CONVERSATIONAL PRACTICE

Issa and Hammond [74] detail the design and implementation of two chatbot systems—KalaamBot and KalimaBot—for Arabic as a foreign language learners. These systems are rule-based rather than LLM-driven, but the pedagogical principles they embody (e.g., dialogic learning, scaffolded interaction) are conceptually aligned with GenAI-assisted education. The study thus offers valuable lessons in chatbot design and learner engagement, indirectly informing future LLM-based implementations.

4) INFRASTRUCTURAL AND POLICY CHALLENGES AFFECTING GENAI ADOPTION IN ARABIC L2 EDUCATION

Hanandeh et al. [75] provide a broader thematic review of artificial intelligence in the Arabic linguistic landscape. While the paper touches upon generative models and language technologies, its scope is wide-ranging, covering traditional AI, NLP infrastructures, and policy considerations. The lack of specific focus on LLM-based tools or empirical engagement with GenAI limits its direct relevance to our review. However, the paper does highlight infrastructural gaps and resource challenges that contextualize the slow adoption of GenAI in Arabic language education.

V. OPPORTUNITIES AND CHALLENGES FOR TEACHING AND LEARNING ARABIC WITH GenAI

GenAI has tremendous potential for transforming language education, particularly Arabic language teaching and learning. Innovative methods that could improve accessibility, improve learning experiences, and meet a range of pedagogical objectives can be introduced by utilizing cutting-edge AI technologies. GenAI may facilitate dynamic engagement, enable individualized learning pathways, and create new opportunities for teacher-student collaboration. However, alongside these opportunities come notable challenges that require attention and thoughtful consideration, such as, privacy, bias, effects on human agency and more. Additionally, the integration of GenAI in language education must address technical, cultural, and ethical complexities, ensuring that the tools developed are effective, inclusive, and respectful of the language's unique characteristics.

A. OPPORTUNITIES PRESENTED BY GENAI

GenAI presents transformative opportunities for language learning and teaching, particularly in addressing challenges associated with low-resource languages such as Arabic. The integration of AI technologies into language education has the potential to democratize access to high-quality learning resources, offer unprecedented levels of personalization, and enhance the engagement and efficacy of language instruction. This section outlines key opportunities presented by GenAI, emphasizing its ability to revolutionize language pedagogy.

1) PERSONALIZED LEARNING EXPERIENCES AND ADAPTIVE FEEDBACK

GenAI can create tailored language learning experiences by adapting content to the learner's proficiency level, dialect preference, and even individual learning styles. By leveraging advanced NLP techniques, AI models such as GPT can analyze user performance and provide adaptive feedback, ensuring that learners progress at an optimal pace. This personalized approach enhances both the effectiveness and engagement of the learning process [76]. Moreover, personalized systems can motivate learners by delivering content that resonates with their specific interests, fostering a deeper connection to the material.

2) AUTOMATIC TEXT GENERATION AND LANGUAGE PRACTICE

Language learners often require diverse opportunities to practice writing and conversational skills. GenAI supports this need by generating context-specific text, designing realistic conversational scenarios, and simulating dialogues. These applications enable learners to practice both writing and speaking skills in a safe, controlled environment. For instance, AI-powered chatbots can engage users in interactive exercises, while generated texts can introduce learners to nuanced expressions and idiomatic usage [77]. Such dynamic practice opportunities bridge the gap between theoretical knowledge and practical application, helping learners achieve fluency faster.

3) DIALECT-SPECIFIC LANGUAGE LEARNING TOOLS

One of the persistent challenges in Arabic language learning is the diversity of dialects, each with unique linguistic features and usage contexts. GenAI models have the potential to address these variations by developing specialized tools tailored to different Arabic dialects. These tools can offer contextually relevant learning experiences, enabling users to engage with content in their preferred dialect. For example, dialect-specific chatbots or lesson generators could provide practical support for learners aiming to master both MSA and regional vernaculars [78]. By bridging the gap between MSA and colloquial Arabic, GenAI tools can significantly enhance the accessibility and relevance of Arabic language education.

4) ADVANCING PRONUNCIATION AND PHONETICS TRAINING

Effective pronunciation and phonetic training are critical components of language learning, as they facilitate clear and accurate communication. GenAI-driven tools, such as speech synthesis and recognition systems, can analyze learners' pronunciation and provide real-time feedback. These systems go beyond merely identifying errors; they guide learners toward correct articulation and help refine subtle phonetic nuances. This is especially important for Arabic, where complex paralinguistic and phonetic features—such as emphatic sounds and regional differences—create difficulties for non-native learners. By integrating advanced speech processing capabilities, AI models can account for Arabic's unique phonetic intricacies and deliver tailored support [79]. Additionally, the development of region-specific AI models would ensure greater inclusivity and accuracy, addressing the needs of diverse learner populations.

5) FACILITATING LOW-RESOURCE LANGUAGE DEVELOPMENT

Low-resource languages often face significant barriers in the availability of teaching materials and technological support. GenAI offers an opportunity to fill these gaps by creating educational content for underrepresented languages and dialects. This is particularly relevant for Arabic, where certain dialects and linguistic features lack sufficient documentation or resources. By leveraging multilingual AI models and transfer learning techniques, it is possible to develop resources that cater to these underserved areas, fostering linguistic diversity and preservation [80].

6) SUPPORTING TEACHER-LEARNER COLLABORATION

While GenAI holds great potential for independent learning, its true value lies in complementing traditional teaching practices. AI-powered tools can support educators by automating repetitive tasks such as grading, generating lesson plans, and curating personalized learning materials. This allows teachers to focus on higher-order instructional activities, such as fostering critical thinking and facilitating discussions. Additionally, collaborative GenAI tools can enable richer interactions between teachers and learners, empowering educators to provide tailored guidance and feedback [81].

7) ENHANCING ENGAGEMENT THROUGH GAMIFICATION AND INTERACTIVE EXPERIENCES

Gamification and interactive learning experiences can be effective in sustaining learner motivation [82]. GenAI can enhance these strategies by creating engaging scenarios, quizzes, and storytelling elements tailored to the learner's progress. For example, AI-generated games could simulate real-world situations, allowing learners to practice language skills in immersive contexts. Such interactive experiences not only make learning enjoyable but also reinforce language acquisition by engaging multiple cognitive pathways [83].

B. CHALLENGES RELATED TO ARABIC LANGUAGE LEARNING

1) HANDLING GENERATIONAL GAP

The generational gap in Arabic language learning reflects differences in technological awareness between learners (Millennials, Gen Z, Gen Alpha) and teachers (Boomers, Gen X). While students favor interactive, gamified methods, teachers often prefer traditional approaches, leading to disengagement and reluctance to adopt new tools [84]. The DigCompEdu framework helps bridge this gap by enhancing teachers' digital skills and fostering tech-driven learning environments. Tools like digital storytelling and speech recognition can align traditional teaching with modern learner preferences [85].

2) DATA SCARCITY AND THE LACK OF COMPREHENSIVE DATASETS

The limited availability of annotated Arabic language datasets poses a significant barrier to training effective GenAI models for Arabic language education [15]. This scarcity is particularly problematic for models that must handle diverse dialects and complex grammatical structures.

3) COMPLEX MORPHOLOGY AND SYNTAX IN THE ARABIC LANGUAGE

Arabic's rich morphology and syntax—characterized by inflectional patterns, root-based word formation, and complex subject-verb-object agreement [31]—pose significant challenges for GenAI, requiring advanced algorithms for accurate text generation and grammar correction. These challenges can be further explored by drawing insights from other morphologically rich languages, such as French and Italian, which address inflectional complexities. Additionally, German, with its complex morphology, offers valuable insights into handling similar linguistic challenges effectively [86], [87].

4) ORTHOGRAPHIC CHALLENGES

Arabic's orthography presents significant challenges for GenAI, driven by its right-to-left script, omission of short vowels represented by diacritics, and positional variations in letter shapes. These features complicate text generation, recognition, and grammar correction. For AI to effectively teach Arabic, it must address these orthographic complexities by producing clear, well-structured text and offering accurate feedback on word formation [31]. Lessons from languages like Chinese, which resolved orthographic challenges through neural networks and annotated datasets, provide useful models for enhancing Arabic language AI applications [88].

5) HANDLING DIALECTAL DIVERSITY IN ARABIC

Developing AI tools that can accurately process the diversity of Arabic dialects remains a significant challenge due to the language's extensive regional variation [89].

Systems trained primarily on MSA often perform poorly with dialectal input unless supplemented with additional data and careful fine-tuning. This limitation is particularly problematic in educational settings, where learners may wish to practice both MSA and their chosen dialect in parallel.

An additional concern with LLMs such as ChatGPT is the imbalance of dialectal representation in training data. Egyptian Arabic, which dominates much of the available online media and digital content, frequently shapes the model's output, leading to responses that disproportionately reflect Egyptian usage. In some cases, this influence even carries over into MSA, subtly altering syntactic structures and lexical choices and thereby compromising the neutrality expected of the formal register. The resulting hybridized variety risks confusing learners and misrepresenting linguistic norms. Addressing these issues requires the development of more representative Arabic corpora that provide balanced coverage of dialects while safeguarding the integrity of MSA in educational applications.

VI. ETHICAL IMPLICATIONS, AND FUTURE DIRECTIONS

A. ETHICAL CONSIDERATIONS IN USING GENAI FOR LEARNING LEARNING

The use of GenAI in language learning raises ethical concerns around privacy, bias, and human agency, as shown in Table 9. This section explores three key areas—Privacy and Data Security, Bias and Fairness, and Impact on Human Agency and Teachers—addressed in dedicated subsections that examine data risks, biased outputs, and AI's influence on learner autonomy and educators' roles, ensuring responsible and equitable GenAI use in education.

1) PRIVACY AND DATA SECURITY CONCERNS

Language learning tools frequently collect user data, raising significant privacy and security concerns, particularly in educational settings [27]. While this data collection enhances user experience by personalizing content and tracking progress, it also introduces ethical and security risks. Without strict safeguards, sensitive data may be vulnerable to unauthorized access, exploitation, or commercial misuse, eroding user trust. Clear, transparent data policies are essential to protect privacy and ensure the benefits of these tools do not compromise user rights.

Protecting user data in AI-driven language learning platforms requires strategies like data anonymization and secure handling practices [90]. Anonymization removes personally identifiable information, ensuring data cannot be traced back to individuals. Secure practices, including encryption, safe storage, and strict access controls, further shield data from breaches and cyberattacks. Encryption protects data during transmission and storage, while access controls limit exposure to authorized personnel only. Adhering to established data privacy regulations is vital to respecting user rights and maintaining trust [91].

2) BIAS AND FAIRNESS IN AI LANGUAGE MODELS

AI models trained on biased datasets risk reinforcing stereotypes, affecting the learning experience and propagating cultural or linguistic biases [15]. Such biases often stem from underrepresentation or social preconceptions embedded in the training data. As a result, generated content may perpetuate negative stereotypes, misrepresenting languages and cultures. In language learning, biased AI models can narrow students' perspectives and reinforce inaccurate cultural or linguistic assumptions. Ensuring AI models are trained on diverse, representative datasets is crucial to minimizing bias and promoting fair, accurate content.

For low-resource languages like Arabic, the limited availability of high-quality data can exacerbate biases in AI models, making it more challenging to develop fair and inclusive language learning tools [8]. Because the training data may overrepresent particular dialects, styles, or settings, this lack of data not only makes it more difficult for AI systems to fully understand language details but also raises the possibility of biases. To address these issues, targeted efforts are needed to collect and curate diverse, high-quality datasets, alongside the development of algorithms designed to mitigate biases and ensure fairness in language learning applications.

3) IMPACT ON HUMAN AGENCY AND THE ROLE OF TEACHERS

While AI can complement traditional teaching, there is a risk that over-reliance on AI may undermine the role of human teachers. Maintaining a balance between AI assistance and human instruction is crucial to preserving the pedagogical value of teacher-student interactions [4]. Maintaining a balance between AI assistance and human instruction is therefore crucial to preserving the pedagogical value of teacher-student interactions.

The integration of AI in language learning raises questions about the evolving roles of teachers and the potential impact on teacher-student relationships [8]. Teachers are essential in creating a safe and stimulating learning environment, providing individualized instruction, and addressing the social and emotional components of learning that AI systems are insufficiently equipped to manage. The overuse or misuse of AI in education risks diminishing these human-centric aspects, leading to a more transactional and less holistic learning experience. In addition, students might become overly dependent on AI tools, disrupting the human-centered nature of education and the teacher-student social dynamics.

4) SAFETY AND CORRECTNESS OF GENAI-GENERATED CONTENT

The safety and correctness of content generated by GenAI tools are fundamental concerns in language learning applications. GenAI models, including large-scale neural networks, are prone to generating outputs that may contain factual inaccuracies, inconsistencies, or unintended biases. Since

TABLE 9. Common pitfalls of using GenAI for language learning.

Pitfall	Description	Example/Implication	Suggested Mitigation
Bias in Language Models	GenAI models may contain biases related to gender, race, culture, or socioeconomic status, which can appear in generated outputs.	Learners could be exposed to stereotypical language or biased views, especially problematic in culturally nuanced languages like Arabic.	Curate training data carefully and implement bias mitigation techniques. Regularly audit outputs for unintended biases.
Overreliance on Translation Accuracy	GenAI translations can appear accurate but may misinterpret context, especially for idioms or nuanced language.	Arabic idiomatic expressions might be misinterpreted, leading to incorrect usage or misunderstandings.	Encourage learners to cross-reference with reliable sources and emphasize contextual awareness in translations.
Lack of Cultural Sensitivity	GenAI may lack understanding of cultural norms, which is critical for meaningful language learning.	Misleading or inappropriate outputs could confuse learners, especially in Arabic, a highly contextual language where meaning heavily relies on cultural cues.	Integrate cultural learning modules alongside AI tools and train models on culturally diverse datasets.
Inconsistent Pronunciation Guidance	AI may struggle to provide accurate pronunciation guidance, especially for language-specific sounds.	Mispronunciation of unique Arabic sounds like 'ayn can lead to incorrect language acquisition.	Use specialized pronunciation tools and incorporate native speaker audio examples for better accuracy.
Privacy Concerns with Sensitive Data	Interactive GenAI tools often collect user data, raising concerns over privacy and data security.	Personal information from learners could be exposed if not managed carefully, especially in educational settings.	Ensure AI tools comply with data privacy standards and provide learners with control over their data sharing preferences.
Dependence on Standardized Language	GenAI tools typically perform better with standardized language, limiting dialectal flexibility.	Learners may lack exposure to regional Arabic dialects, impacting real-world communication skills.	Integrate dialect (modules) in the curriculum and expose learners to various Arabic dialects for a balanced learning experience.
Limited Customization for Learning Styles	GenAI may not fully adapt to diverse learning needs or levels of proficiency.	Content might be too advanced or too simplistic, reducing its effectiveness for individual learners.	Implement adjustable difficulty levels and allow customization based on learner preferences and proficiency.
Risk of Plagiarism and Overuse of AI Assistance	Heavy reliance on AI can lead to reduced active learning and superficial understanding.	Learners may rely on AI for writing tasks, reducing opportunities for original expression and deeper engagement.	Emphasize active learning activities and balance AI assistance with human-driven exercises.
Challenges in Developing Critical Thinking	AI-provided answers may discourage learners from engaging in deeper exploration or critical analysis.	Learners might accept AI-generated explanations passively rather than developing their own understanding.	Design tasks that require critical thinking and reflection, encouraging learners to question and analyze AI outputs.

these models rely on probabilistic methods and extensive datasets scraped from the internet, they may inadvertently produce misleading or inappropriate content, which can negatively impact learners. Moreover, the opaque nature of deep learning models makes it difficult to fully understand how decisions are made, further complicating the verification of content correctness. To address these issues, researchers have proposed incorporating human oversight mechanisms, such as expert review and reinforcement learning with human feedback (RLHF), to improve output reliability. Additionally, adversarial testing and dataset curation techniques are being explored to refine training data and minimize the risk of generating unsafe content [92]. Future research should emphasize designing GenAI models with built-in content verification systems and increased transparency to ensure that AI-assisted language learning remains practical, ethical, and free from harmful misinformation.

5) SUPPORTING ISLAMIC EDUCATION WITH GENAI

GenAI offers valuable support for Islamic education by enhancing access to Arabic texts of the Quran and Hadith, foundational to Islamic learning [93]. However, Islamic pedagogy emphasizes Isnad—the chain of transmission—and the necessity of direct learning from human teachers to preserve trusted knowledge networks. GenAI should therefore serve as an instrumental tool, improving accessibility and fluency without diminishing the relational, human-centered core of Islamic education. The complexity of using technology to engage with religious texts is further heightened by the Arabic language's inherent ambiguity and conciseness, which can challenge even advanced AI models in capturing nuanced meanings [93]. This highlights the importance of AI as a complementary aid for research and understanding, reinforcing the interpretive role of human scholars to ensure accuracy and contextual depth through oversight.

B. OPEN ISSUES AND FUTURE WORK

Addressing the unique challenges of using GenAI for Arabic language learning requires a multi-faceted approach spanning the following areas.

1) DEVELOPING COMPREHENSIVE ARABIC DATASETS

Building robust datasets for Arabic remains a critical open issue due to the language's diverse dialects and complex morphology. Community-driven approaches, such as crowd-sourcing and community-based data collection, can play a significant role in expanding available data resources [78]. These approaches allow for creating more representative datasets that capture the full spectrum of language use across different regions and contexts. Additionally, collaborating with educational institutions and linguists can ensure dataset quality and comprehensiveness by drawing on academic expertise [89]. Engaging experts in Arabic linguistics helps ensure that the datasets reflect the richness of the language, including variations in syntax, semantics, and phonetics. These collaborative efforts are essential for training GenAI models that can cater to the nuanced demands of Arabic language learners. Furthermore, incorporating diverse perspectives during data collection ensures that the datasets account for different cultural, social, and educational contexts, making AI-driven tools more relevant and effective.

2) IMPROVING GenAI MODELS FOR ARABIC

GenAI models for Arabic can benefit from techniques like transfer learning and multilingual modeling, leveraging knowledge from high-resource languages to enhance

performance in low-resource contexts [15]. By transferring learned features from languages like English, these methods help address the scarcity of high-quality Arabic data. Ongoing research in morphological analysis and dialectal processing is essential to manage Arabic's complex grammar and regional variations [94]. Arabic's root-based morphology and intricate inflection system demand specialized models capable of accurate analysis and generation. The diversity of dialects across the Arab world, which often diverge significantly from MSA and each other, poses further challenges. Effective models must process both formal and informal language variations. Advancing these capabilities is crucial to developing GenAI models that are not only more accurate but also culturally and linguistically relevant.

3) EXPLORING MULTIMODAL AI FOR LANGUAGE LEARNING

Integrating text, audio, and video through multimodal AI can enrich the language learning experience, offering a more comprehensive approach to skill development [70]. Specifically, combining text, audio, and video inputs provides diverse ways to practice and reinforce language skills, while speech recognition and text-to-speech technologies facilitate spoken language practice and pronunciation feedback, which are crucial for oral proficiency in Arabic [59]. In addition, expanding the scope of multimodal AI could include kinesthetic learning through touch, fine motor skills, and even sensory inputs such as taste and smell. Although still exploratory, incorporating these modalities into AI systems could create immersive, holistic language learning by integrating auditory, kinesthetic, and sensory feedback.

4) DEVELOPING AI-DRIVEN ARABIC ASSESSMENT TOOLS

Future research on GenAI for Arabic language assessment must address challenges related to evaluating all language skills—speaking, listening, reading, and writing—while accommodating Arabic's complex morphology, multiglossia, and dialectal diversity. Developing adaptive assessments that accurately reflect proficiency across MSA and regional dialects remains a key area for improvement. Enhancing pronunciation and fluency evaluation, expanding datasets to include underrepresented dialects, and refining real-time feedback systems are essential to creating inclusive and culturally relevant assessments. Future work should also focus on mitigating bias and ensuring fairness, requiring collaborative efforts between linguists, educators, and AI researchers to advance more accurate and equitable Arabic language assessment tools.

5) INTEGRATING AI WITH TRADITIONAL TEACHING METHODS

A promising way to enhance GenAI's impact in language learning is through blended learning, integrating AI tools with traditional instruction to create hybrid environments that combine AI capabilities with human guidance [4]. This model personalizes learning by using AI to provide immediate feedback, track progress, and offer tailored exercises, while traditional instruction fosters deeper understanding, communication skills, and critical thinking. Blended learning also offers flexibility, allowing students to learn at their own pace while benefiting from face-to-face interaction. Incorporating AI-driven tools into Arabic language education can modernize curricula, making lessons more dynamic and adaptable to diverse learning styles. This integration enhances engagement, accessibility, and effectiveness, ultimately supporting greater proficiency and fluency.

6) ETHICAL AI POLICY GUIDELINES FOR LANGUAGE EDUCATION

To ensure responsible AI deployment, it is crucial to establish clear guidelines for developers and educators that address user privacy, fairness, and accessibility, alongside regulatory frameworks that safeguard students' rights and promote equitable access to educational resources [15], [57]. These guidelines should focus on protecting sensitive data, ensuring that AI systems do not exploit or misuse personal information, and establishing robust safeguards to prevent unauthorized access or breaches. Moreover, fairness should be a central consideration, ensuring that AI systems are designed to avoid biased outcomes that could adversely affect certain student groups. Furthermore, accessibility is critical, as AI tools must be available and usable for all students, regardless of their backgrounds or abilities. Developers and educators should prioritize privacy, fairness, and accessibility, acknowledging that students in underserved areas may face barriers to fully engaging with AI-driven tools.

7) LEVERAGING AI FOR OTHER LOW-RESOURCE LANGUAGES

Advancements from applying GenAI to Arabic can guide efforts for other low-resource languages, enabling cross-linguistic knowledge transfer and fostering multilingual AI model development [58], [77]. This promotes inclusive, efficient AI-driven language learning tools beyond high-resource languages. Lessons learned from Arabic can help improve AI technologies for languages with limited datasets, creating scalable frameworks for future applications. Multilingual models encourage collaboration among language experts, educators, and AI researchers, enhancing understanding of shared and unique linguistic features. This approach not only strengthens AI tools for underrepresented languages but also expands language education, supporting greater linguistic diversity and equitable access to learning worldwide.

VII. CONCLUSION

This paper has reviewed the potential of Generative AI (GenAI) in Arabic language learning, emphasizing the unique opportunities and challenges it presents. Through the taxonomy of applications we developed, this review contributes to the state of knowledge by systematically categorizing how GenAI, particularly Large Language Models (LLMs), can

enhance Arabic learners' skills in speaking, writing, and comprehension, addressing the language's distinct dialectal and structural complexities. By drawing insights from English language applications, we identified pathways to adapt GenAI specifically to Arabic's multiglossic needs, supporting both linguistic proficiency and cultural competence. This review highlights key challenges for responsible GenAI implementation in Arabic education, including developing adaptive systems to reflect Arabic's linguistic diversity and addressing ethical concerns around data privacy, bias, and accessibility for low-resource languages. By outlining these challenges, this paper not only provides a foundation for future research but also offers actionable insights for educators, researchers, and developers. Finally, the importance of interdisciplinary collaboration among AI, linguistics, and education experts is underscored, as such partnerships are essential for creating culturally sensitive and effective GenAI tools.

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